

Yibo Yin

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EDUCATION	École polytechnique fédérale de Lausanne (EPFL) <i>M.Sc. in Digital Humanities</i> • Research Focus: Computer Graphics, Computer Vision, and Machine Learning • GPA: 5.56 /6.00	Lausanne, Switzerland 09/2025 - Present
	Wuhan University <i>B.Eng. in Computer Science and Technology</i> • GPA: 3.93 /4.00 (top 4%)	Wuhan, China 09/2021 - 06/2025
PUBLICATIONS	MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization Ningna Wang, Rui Xu, Yibo Yin , Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo <i>ACM SIGGRAPH Asia 2025 (Conference Track)</i>	
RESEARCH EXPERIENCE	Image and Visual Representation Lab, EPFL <i>Research Assistant Advisor: Prof. Sabine Süsstrunk ✉ and Dr. Zhuoqian Yang ✉</i> • Working on Weight Space Representation and Implicit Neural Representations.	06/2026 - Present
	Realistic Graphics Lab, EPFL <i>Research Assistant Advisor: Prof. Wenzel Jakob ✉</i> • Implemented differentiable Gaussian Process Implicit Surface (GPIS) sampling with Dr.Jit. • Developed a hybrid surface – volume rendering approach for Many-Worlds inverse rendering.	03/2026 - 06/2026
	Computer Graphics Lab, The University of Texas at Dallas <i>Research Assistant (Remote) Advisor: Prof. Xiaohu Guo ✉</i> • Extended the evaluation algorithm in the SIGGRAPH Course Note paper <i>Evaluation of Loop Subdivision Surfaces</i> and the SGP paper <i>Fitting Sharp Features with Loop Subdivision Surfaces</i> to structured medial axis meshes under Loop Subdivision. • Extended the simplification algorithm in the SIGGRAPH paper <i>Q-MAT: Computing Medial Axis Transform by Quadratic Error Minimization</i> to structured medial axis meshes. • Proposed a subdivision surface fitting algorithm in collaboration with advisor, designed to preserve the sharp features of the structured medial axis mesh during fitting.	04/2024 - 11/2025
	Waterloo Computer Graphics Lab, University of Waterloo <i>Research Assistant (Remote) Advisor: Prof. Toshiya Hachisuka ✉</i> • Reproduced the algorithm in the SIGGRAPH paper <i>A Practical Walk-on-Boundary Method for Boundary Value Problems</i> for solving the boundary value problem of Laplace's equation with Dirichlet boundaries.	06/2024 - 08/2024
	Graphics and Vision Lab, Wuhan University <i>Research Assistant Advisor: Prof. Chunxia Xiao ✉</i> • Assembled a set of equipment with an Intel depth camera and STMicroelectronics ToF sensors, creating a dataset of over 10,000 images and corresponding sensor data. • Proposed a novel monocular depth estimation method using RGB images and sensor data as input.	05/2023 - 01/2024
SELECTED PROJECTS	Physically Based Renderer <i>CS-440 Advanced Computer Graphics Rendering Competition, EPFL</i> • Extended the Nori framework with heterogeneous volumetric path tracing, delta and ratio tracking, Henyey-Greenstein phase functions, and MIS over emitter, phase, and BSDF sampling. • Implemented HDR environment map lighting, thin-lens depth of field, motion blur, principled BSDF, and GGX rough dielectric materials for a storm-scene rendering pipeline.	2026

AWARDS

- **Outstanding Student** (10% school-wide), Wuhan University 2022, 2023, 2024
- **Second Class Scholarship** (10% school-wide), Wuhan University 2022
- **Third Class Scholarship** (15% school-wide), Wuhan University 2023, 2024
- **Lei Jun Computer Innovation and Development Fund**, Wuhan University 2024

TECHNICAL SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT 102)
- **Programming Languages:** C++, C, Python, C#, GLSL, SQL, Verilog HDL, Java, $\text{\LaTeX}$
- **Libraries / Frameworks / Tools:** PyTorch, Dr.Jit, Mitsuba, PBRT-v3, Blender, Eigen, CGAL, libigl, Git, CMake